

INDUSTRIAL PRODUCT DATASHEET

XIAO Vision AI Camera

This XIAO gadget is a vision solution that combines the Grove Vision AI V2 module, XIAO ESP32-C3 controller, and an OV5647 camera, housed in a custom 3D-printed enclosure.

SKU

104990982

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PRICE

\$24.90(Excl. VAT)



Overview



This XIAO gadget is a vision solution that combines the Grove Vision AI V2 module, XIAO ESP32-C3 controller, and an OV5647 camera, housed in a custom 3D-printed enclosure. The Grove Vision AI V2 module features the Himax WiseEye2 platform, integrating an Arm Cortex-M55 processor and Ethos-U55 NPU for edge AI computing. The camera supports TensorFlow and PyTorch frameworks and is compatible with Arduino IDE. It offers no-code model deployment and visualization of inference results through SenseCraft AI.

Key Features

01

Feature 1

Powerful AI Processing Capabilities

02

Feature 2

Versatile No-Code AI Model Deployment & Visualization

03

Feature 3

Wi-Fi Enabled Smart Vision

04

Feature 4

Local Closed-Loop Automation with Home Assistant Integration

05

Feature 5

Fully Open Source

06

How XIAO Vision AI Camera Works?

Product Gallery





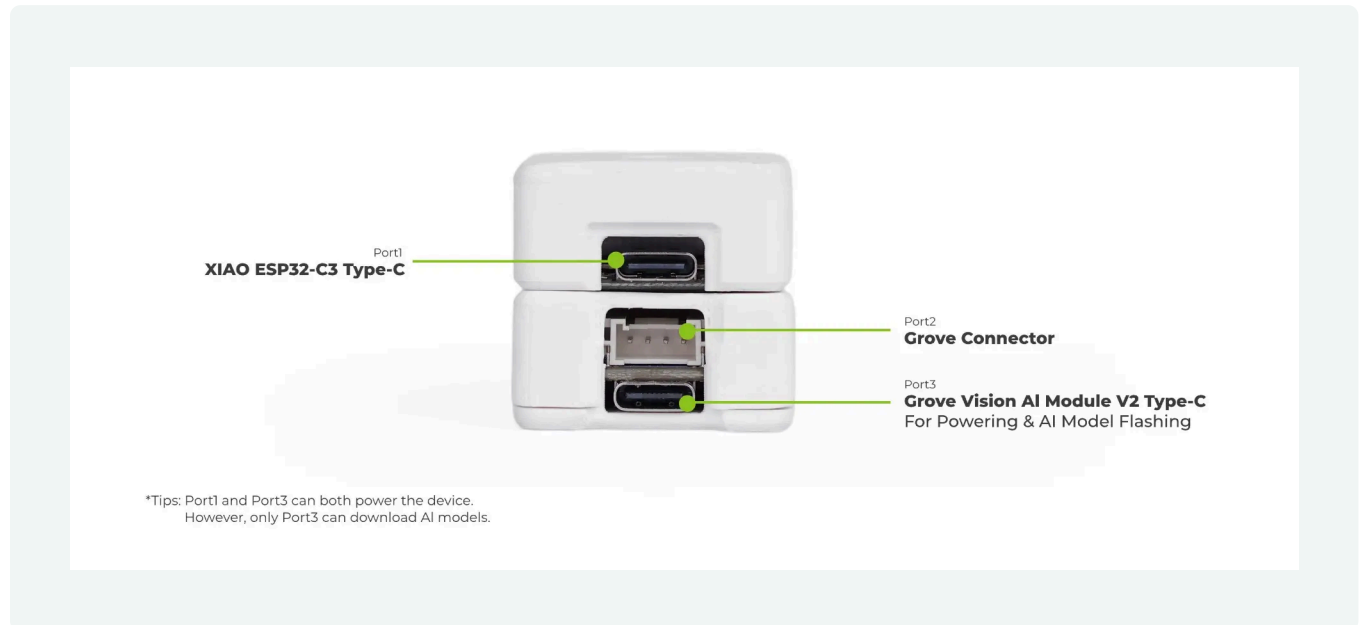
Specification

XIAO ESP32-C3

for remix on Thingiverse.

Parameter	Description	
Camera	Sensor	OmniVision OV5647
Still Image Resolution	2592 × 1944 (5 MP)	
Field of View	62°	
Video Modes	1080p @ 30 fps; 720p @ 60 fps	
Pixel Size	1.4 μm × 1.4 μm	
Focal Length	3.4 mm (adjustable)	
Aperture (F-Stop)	F/2.8	
CMOS Size	1/4"	
Microcontrollers	Master Controller	● 2.4GHz Wi-Fi/Bluetooth 5.0● RISC-V single-core 32-bit chip processor with a four-stage pipeline that operates at up to 160 MHz● 400KB SRAM & 4MB Flash
Imaging Processing UnitGroveVision AI Module V2, powered byHimax WiseEye2 HX6538	● dual-core Arm Cortex-M55 @ 400 MHz & @ 150 MHz● Ethos-U55 microNPU @ 400 MHz ● Configurable SRAM up to 2432 KB; 64 KB Boot ROM● CSI camera connector● Grove I ² C/UART/SPI● Micro SD card slot (DS mode, up to 25 MHz SDIO)● PDM microphone; micro SD card slot	
Power Supply	5 V via USB Type-C (XIAO ESP32-C3)	
Baud Rate	115200 bps	
Operating Temperature	-20 °C to 70 °C	
Enclosure	Dimension: 31 x 49 x 32mmMaterial: PLAColor: WhiteFabrication: 3D PrintingAlso open sourced	

Hardware Overview



Applications

- **Industrial automation:** quality inspection, predictive maintenance, voice control, etc.
- **Smart city:** equipment monitoring, energy management, etc.
- **Transportation:** status monitoring, location tracking, etc.
- **Intelligent agriculture:** environmental monitoring, etc.
- **Mobile IoT devices:** wearable devices, handheld devices, etc.

Part List

XIAO Vision AI Camera	×1
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Compliance & Logistics

HSCODE	9031809090
USHSCODE	9031808085
EUHSCODE	9013101000
COO	CHINA

Logistics	Value
ECCN	EAR99
HSCODE	9031809090
USHSCODE	9031808085
EUHSCODE	9013101000
COO	CHINA

Resource Links

- <https://www.seeedstudio.com/XIAO-Vision-AI-Camera-p-6450.html>
- [3D Printed Foldable Holder for Grove Vision AI Module V2 Kit \(Grove Vision AI Module V2 + OV5647 Camera + XIAO\)](#)
- [HX6538 Datasheet](#)
- [Himax examples for Seeed Grove Vision AI Module V2](#)

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